

A Survey of Diffusion and Flow-Based Models for Controllable Image Generation

InteractiveSurvey

Abstract

Generative modeling has undergone a paradigm shift with the advent of diffusion and flow-based models, yet their reliance on iterative sampling often incurs prohibitive computational costs for real-time applications. Recent theoretical advancements suggest that unifying these paradigms through probability flow ordinary differential equations can reconcile training stability with sampling efficiency. This survey critically examines the intersection of diffusion and flow-based architectures, specifically focusing on their convergence to enable controllable image generation. We synthesize current research bridging these fields by analyzing how flow matching techniques optimize the underlying transport maps to straighten generative trajectories. Our review establishes the mathematical equivalence between stochastic denoising and deterministic velocity regression, demonstrating how optimal transport principles significantly reduce integration steps while preserving high-fidelity synthesis. Furthermore, we provide a systematic categorization of advanced numerical solvers, adaptive step-size controllers, and manifold extensions that address numerical instability and enhance inference speed. By mapping the correspondences between score matching and flow matching objectives, this work offers practitioners a structured framework for navigating efficiency-quality trade-offs in modern generative systems. Ultimately, this paper serves as a foundational reference for developing the next generation of fast, controllable, and theoretically grounded image synthesis models.

1 Introduction

Generative modeling has undergone a paradigm shift with the advent of diffusion models, which have established new benchmarks in image synthesis quality and diversity. By iteratively de-

noising data through stochastic differential equations, these models effectively capture complex high-dimensional distributions. However, the reliance on numerous iterative steps for sampling often incurs significant computational costs, limiting real-time applications. Concurrently, flow-based generative models have emerged as a compelling alternative, offering deterministic trajectories and exact likelihood evaluation. The convergence of these two fields suggests a unified theoretical framework where stochastic diffusion processes can be reinterpreted as deterministic flows, promising to reconcile the training stability of diffusion with the sampling efficiency of normalizing flows.

This survey paper critically examines the intersection of diffusion and flow-based models, specifically focusing on their application to controllable image generation. While existing literature often treats these architectures as distinct entities, recent theoretical advancements reveal that they share a common foundation in probability flow ordinary differential equations. The primary objective of this work is to synthesize current research that bridges these paradigms, analyzing how flow matching techniques can optimize the transport maps underlying diffusion processes [1]. By exploring the equivalence between stochastic denoising and deterministic velocity regression, this review aims to clarify how hybrid approaches can achieve superior control over generative trajectories while minimizing inference latency.

The discussion begins by establishing the theoretical foundations of flow matching and its mathematical equivalence to diffusion dynamics. It details how the probability flow ODE serves as a deterministic counterpart to stochastic diffusion, allowing for the direct parameterization of vector fields without explicit score estimation. Furthermore, the integration of optimal transport theory is explored as a mechanism for constructing straight

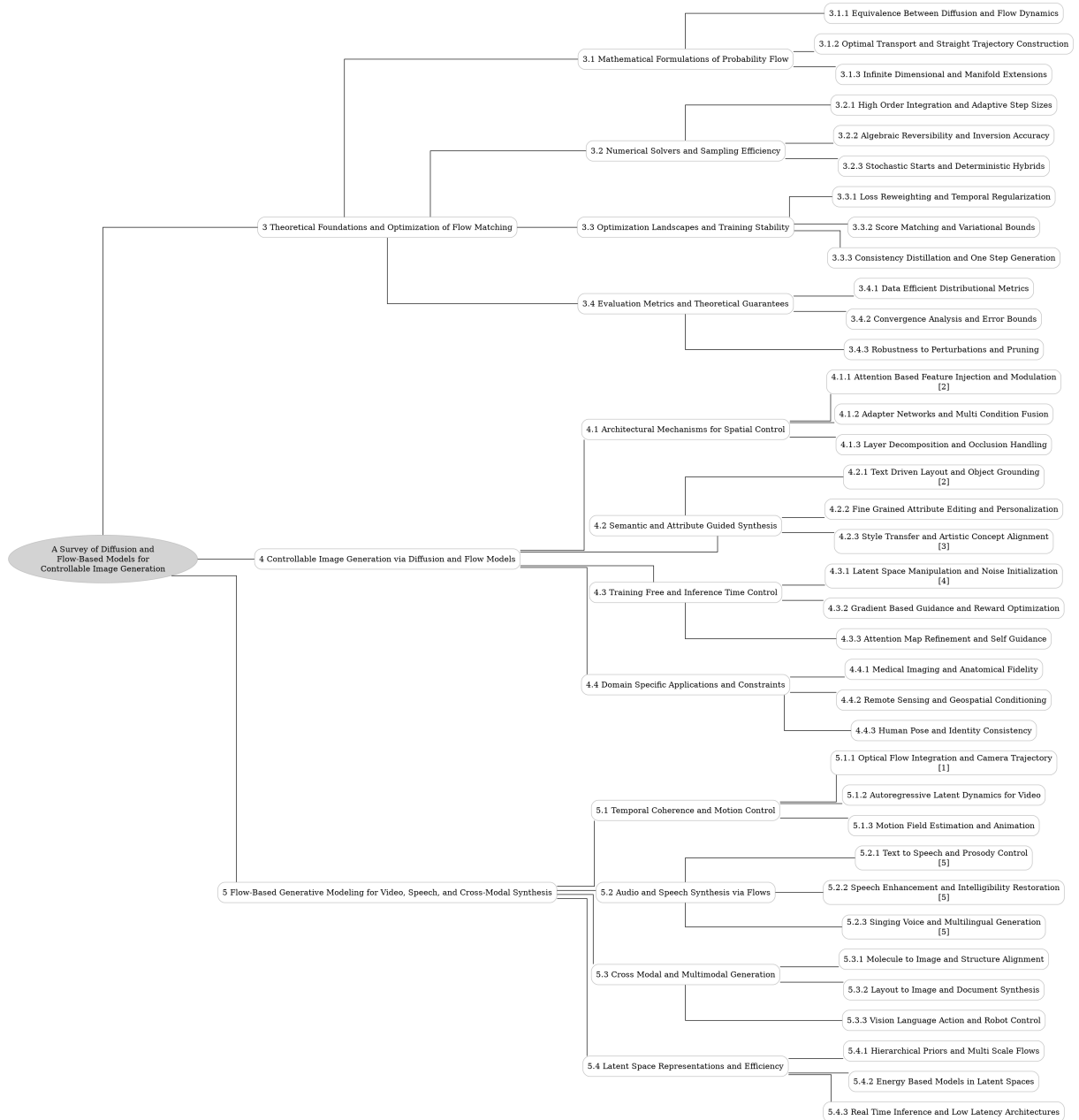


Figure 1: The outline of our survey: A Survey of Diffusion and Flow-Based Models for Controllable Image Generation

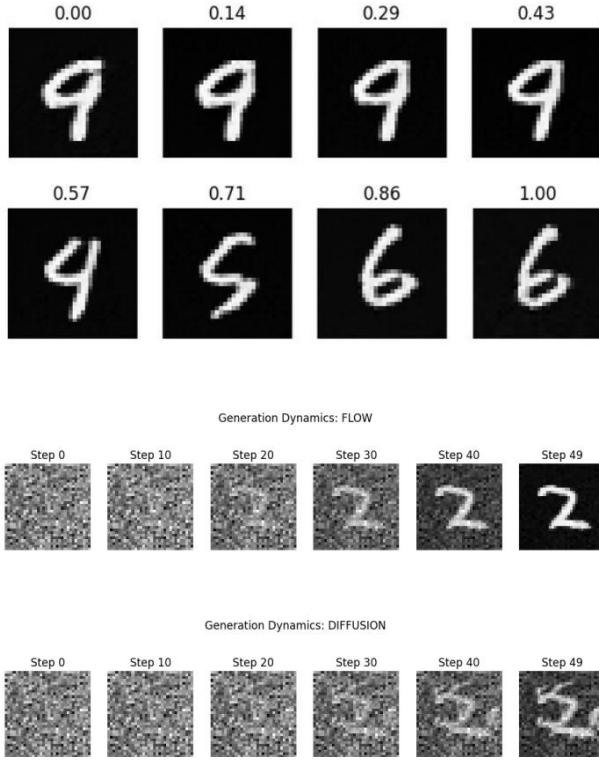


Figure 2: Chart from 'Efficiency vs. Fidelity: A Comparative Analysis of Diffusion Probabilistic Models and Flow Matching on Low-Resource Hardware' (arXiv:2511.19379)

trajectories, which significantly reduces the integration steps required during sampling. This section also addresses the extension of these concepts to infinite-dimensional spaces and Riemannian manifolds, highlighting methods to maintain geometric consistency and prevent numerical instability when generating data on complex topologies.

Subsequently, the survey analyzes numerical solvers and sampling strategies designed to enhance inference efficiency. It contrasts standard first-order Euler methods with high-order integration schemes and adaptive step-size controllers that reduce truncation errors and accelerate convergence. The role of algebraic reversibility is scrutinized, examining the trade-offs between analytical invertibility and architectural expressivity in high-dimensional settings. Additionally, the transition from stochastic starts to deterministic hybrids is discussed, illustrating how reformulating stochastic processes as ordinary differential equations enables faster generation while preserving the robustness of stochastic training objectives through predictor-corrector mechanisms.

Finally, the review investigates optimization

landscapes and training stability, focusing on loss reweighting strategies and temporal regularization techniques. These methods address the varying significance of different timesteps along the probability path, ensuring that learning is biased toward regions critical for generation quality. The interplay between guidance mechanisms and vector field stiffness is also evaluated, providing insights into how modern frameworks mitigate Lipschitz explosions near data manifolds. Together, these components form a comprehensive overview of the algorithmic innovations driving the current state of controllable image generation, setting the stage for future developments in efficient and stable generative modeling.

The primary contribution of this survey is the unification of disparate research threads connecting diffusion and flow-based models into a coherent theoretical and practical framework. Unlike previous reviews that treat these topics in isolation, this work explicitly maps the mathematical correspondences between score matching and flow matching objectives, clarifying how optimal transport principles can be leveraged to straighten generative paths [1]. Moreover, it provides a systematic categorization of numerical solvers and optimization techniques tailored for controllable generation, offering practitioners a structured guide to selecting appropriate architectures for specific efficiency-quality trade-offs. By synthesizing these insights, this paper establishes a foundational reference for advancing the next generation of fast, controllable, and theoretically grounded image synthesis models.

2 Theoretical Foundations and Optimization of Flow Matching

2.1 Mathematical Formulations of Probability Flow

2.1.1 Equivalence Between Diffusion and Flow Dynamics

The theoretical bridge between stochastic diffusion models and deterministic flow-based generative approaches is established through the Probability Flow Ordinary Differential Equation (ODE). While standard diffusion processes simulate Stochastic Differential Equations to transform data into noise and back, it has been demonstrated that for every such stochastic trajectory, there exists a corresponding deterministic ODE that preserves the same marginal probability den-

sities at each time step. This Probability Flow ODE defines a continuous vector field that transports particles from the prior distribution to the data distribution without introducing stochasticity, effectively rectifying the curved stochastic paths into deterministic flows. Consequently, the generative mechanism of diffusion models can be reinterpreted as solving an initial value problem governed by this specific time-dependent velocity field.

Flow Matching formalizes this connection by directly parameterizing and learning the vector field required to solve the continuity equation, bypassing the need to explicitly estimate the score function utilized in traditional denoising diffusion. By constructing a conditional probability path, typically via linear interpolation between data and noise samples, Flow Matching trains a neural network to regress the instantaneous velocity that moves samples along these trajectories. Mathematical analysis reveals that under specific conditions, particularly when utilizing Gaussian conditional paths, the optimal velocity field learned by Flow Matching coincides exactly with the drift term of the Probability Flow ODE derived from diffusion theory. This equivalence implies that both frameworks are optimizing toward the same underlying transport map, albeit through different objective formulations and training conventions.

Furthermore, recent unifying frameworks involving stochastic interpolants generalize both paradigms, showing that diffusion and flow matching represent distinct points within a broader spectrum of generative dynamics defined by the choice of interpolant and noise schedule. The distinction largely resides in the training objective: diffusion models minimize a denoising score matching loss related to the gradient of the log-density, whereas flow matching minimizes a regression loss on the velocity field directly. Despite these algorithmic differences, the resulting generative processes share identical marginal distributions throughout the transformation. This theoretical alignment allows for the interchangeability of architectural components and sampling strategies, enabling practitioners to leverage the straighter trajectories of flow-based models to reduce integration steps while retaining the robust training stability characteristic of diffusion objectives.

2.1.2 Optimal Transport and Straight Trajectory Construction

Optimal Transport (OT) provides a rigorous geometric framework for constructing straight trajectories in generative modeling by minimizing the kinetic energy required to transport mass between source and target distributions. Under the Benamou-Brenier formulation, the optimal velocity field corresponds to the gradient of a convex potential, theoretically yielding straight-line paths that approximate the Wasserstein-2 geodesic. This contrasts sharply with standard diffusion processes, which often follow highly non-linear trajectories due to arbitrary couplings, thereby necessitating numerous iterative steps during inference. By enforcing an OT-based coupling, the resulting ordinary differential equations describe direct transformations, significantly reducing the integration length and improving sampling efficiency.

However, computing the exact optimal transport plan for large-scale datasets is computationally prohibitive due to the super-cubic complexity of solving the global assignment problem. To address this intractability, recent methodologies have shifted towards minibatch approximations, such as Minibatch OT and Minibatch Sinkhorn coupling, which estimate optimal pairings within local subsets of data. These approaches effectively lower the computational burden while preserving the marginal-preserving properties essential for valid probability paths. Furthermore, semi-discrete formulations and entropic regularization techniques have been employed to stabilize the optimization landscape, allowing for the construction of data-informed couplings that closely mimic the global optimal map without requiring full matrix computations.

The practical implementation of these straight flows often involves rectification procedures that iteratively refine the velocity field to align with the optimal transport direction. By optimizing the deviation between the learned vector field and the theoretical straight-line velocity, models can progressively straighten curved trajectories into near-linear paths. This process not only accelerates convergence but also enhances the stability of the generated samples by mitigating the stiffness associated with impulsive control dynamics. Consequently, the integration of optimal transport principles into flow matching frameworks establishes a robust mechanism for achieving high-fidelity gen-

eration with minimal inference steps, bridging the gap between theoretical optimality and practical scalability.

2.1.3 Infinite Dimensional and Manifold Extensions

Extending flow-based generative models beyond finite-dimensional Euclidean spaces requires addressing the intrinsic low-dimensional geometry of high-dimensional data distributions. Theoretical frameworks increasingly model data support as Riemannian manifolds embedded within ambient spaces, where standard vector fields must be replaced by tangent bundle flows to ensure trajectories remain on the learned geometry. This shift necessitates redefining the continuity equation and probability paths using metric tensors, allowing for diffeomorphic mappings that respect the curvature and topology of the underlying data structure rather than assuming flat, isotropic noise distributions.

A critical challenge in these extensions is the structural instability of guidance mechanisms near the data manifold, often manifested as Lipschitz explosion in the score function. As sampling trajectories approach the support of the real distribution, the divergence term within the guided vector field can blow up, causing numerical stiffness and off-manifold deviations. To mitigate this, recent approaches decompose the guidance force into invariant components, employing time-dependent schedules and manifold-preserving projections. These techniques effectively bound conservation-violating terms, ensuring that the reverse process reduces the distance to the condition while maintaining geometric consistency without requiring retraining.

Furthermore, operationalizing these models in infinite-dimensional or latent settings leverages bracket-generating conditions to span tangent spaces with minimal vector fields. By utilizing Lie algebra properties, it is possible to generate expressive transport dynamics using a cardinality of fields independent of the ambient dimension. This facilitates scalable computation on complex topologies, such as those found in medical imaging or 3D generation, where the data manifold may not be diffeomorphic to standard Euclidean space. Consequently, these extensions provide a rigorous foundation for rectifying transport paths, reducing trajectory curvature, and achieving stable generation across diverse modalities.

2.2 Numerical Solvers and Sampling Efficiency

2.2.1 High Order Integration and Adaptive Step Sizes

Standard continuous normalizing flows often rely on first-order Euler solvers, which necessitate fine-grained temporal discretization to maintain accuracy along curved trajectories, thereby inflating the number of function evaluations. To mitigate this inefficiency, high-order integration schemes such as Heun’s method and Runge-Kutta variants are employed to reduce local truncation error, allowing for larger step sizes without compromising sample quality. These methods approximate the integral of the velocity field with higher precision, effectively straightening the learned transport paths and significantly accelerating the inference process compared to basic explicit solvers.

Beyond fixed-step high-order methods, adaptive step-size controllers dynamically adjust the integration interval based on estimated local error bounds, ensuring stability in regions of high vector field stiffness while maximizing step size in smoother domains. This approach is critical for handling the numerical instability inherent in models with complex dynamics, where rigid discretization can lead to divergence or poor convergence. By leveraging embedded error estimators within the solver architecture, the system automatically balances computational cost against approximation fidelity, preventing the global truncation error from accumulating excessively over the integration horizon.

The synergy between high-order schemes and adaptive stepping fundamentally alters the efficiency landscape of flow-based generative modeling. While standard approaches may require hundreds of steps to achieve low Fréchet Inception Distance scores, advanced solvers can often reach comparable performance with an order of magnitude fewer evaluations. However, this comes with increased memory overhead due to the caching of intermediate activations required for higher-order stages and error estimation. Consequently, optimal deployment involves a trade-off between the computational intensity per step and the total reduction in integration steps, particularly when scaling to high-dimensional data distributions.

2.2.2 Algebraic Reversibility and Inversion Accuracy

Algebraic reversibility in generative flows fundamentally relies on the tractable computation of inverse mappings and Jacobian determinants. While autoregressive structures and coupling layers enforce invertibility by design, ensuring analytical inversion often conflicts with model expressivity and scaling. Specifically, the presence of inverse-derivative terms and high-dimensional Jacobians introduces computational bottlenecks that grow prohibitively with data dimensionality. Consequently, many modern approaches sacrifice strict analytical invertibility for architectural flexibility, relying instead on numerical solvers to approximate the reverse trajectory during inference.

The accuracy of these inversion processes is intrinsically linked to the discretization errors inherent in solving the underlying ordinary or stochastic differential equations. Local truncation errors accumulate over integration steps, leading to global deviations that degrade sample fidelity and reconstruction precision. Techniques such as sequential reflow aim to mitigate these discrepancies by straightening probability paths, thereby reducing the stiffness of the vector field and allowing for larger integration step sizes without compromising stability. However, the order of accuracy remains constrained by the solver choice, where higher-order methods like Heun’s scheme offer improved precision at the cost of increased function evaluations compared to Euler discretizations.

Furthermore, inversion accuracy is sensitive to spectral biases and the conditioning of the learned velocity field, particularly in linear inverse problems. Errors propagated during the forward pass can become irreversible in open-loop generation settings, where early hallucinations fix the prior for subsequent steps. To address this, recent formulations explore parameterization-invariant strategies and corrective mechanisms, such as predictor-corrector sampling, which implicitly update the manifold to align with base generative dynamics. Ensuring robust inversion thus requires a delicate balance between enforcing algebraic constraints, minimizing numerical truncation, and maintaining the geometric consistency of the latent space throughout the transformation.

2.2.3 Stochastic Starts and Deterministic Hybrids

Stochastic starts in generative modeling traditionally rely on simulating Stochastic Differential Equations (SDEs) where trajectories evolve from a noise prior through the addition of Brownian motion. While this approach effectively captures complex data manifolds, the inherent randomness necessitates iterative sampling procedures that can be computationally expensive and prone to trajectory divergence. Theoretical analysis reveals that because the construction of intermediate states remains stochastic, the optimal solution for the governing objective functions rarely converges to zero, requiring careful discretization and multiple evaluation steps to approximate the target distribution accurately.

To mitigate these inefficiencies, recent advancements have reformulated these stochastic processes as deterministic Probability Flow Ordinary Differential Equations (ODEs). This transformation preserves the marginal distributions of the original SDE while enabling the use of adaptive, high-order ODE solvers for significantly faster inference. By defining a deterministic vector field that guides samples along straighter paths between the source noise and target data, these methods reduce the number of function evaluations required during generation. The resulting flows offer a continuous-time perspective where the evolution of probability density is governed entirely by a learned drift term, eliminating the variance introduced by stochastic increments.

Deterministic hybrids further extend this paradigm by integrating normalizing flows with diffusion frameworks to create more flexible generative architectures. Instead of relying solely on a fixed Gaussian prior, these models employ flow networks to parameterize data-dependent priors at intermediate timesteps, effectively bridging the gap between stochastic diffusion and deterministic transport. Techniques such as fine-tuning initial conditions or re-establishing formulations based on theoretical relationships between Flow Matching and Score-based models allow for precise trajectory control. These hybrid approaches leverage the robustness of stochastic training objectives while capitalizing on the efficiency and invertibility of deterministic inference, yielding superior sample quality with reduced computational overhead.

2.3 Optimization Landscapes and Training Stability

2.3.1 Loss Reweighting and Temporal Regularization

Loss reweighting strategies are pivotal in stabilizing the training of continuous normalizing flows and flow matching models by addressing the varying significance of different timesteps along the probability path. Rather than treating all temporal intervals equally, recent approaches employ time-dependent weighting schemes to bias learning toward specific regions, such as the mid-trajectory or endpoints, where approximation errors most critically impact generation quality. These schemes effectively transform the objective into differently weighted evidence lower bound formulations, allowing practitioners to prioritize accurate denoising in high-curvature regions while mitigating the dominance of primary velocity components that can obscure guidance signals in later iterations.

Complementing explicit reweighting, temporal regularization techniques impose structural constraints on the learned vector field to ensure smooth and efficient transport maps. By incorporating optimal transport principles, methods such as kinetic and Jacobian regularization penalize excessive curvature and rapid changes in the flow, encouraging straighter trajectories that reduce integration steps during inference. This regularization limits the total movement between density distributions at consecutive iterations, selecting transport maps that minimize the Wasserstein distance while maintaining fidelity to the target distribution, thereby enhancing numerical robustness and preventing posterior collapse in latent variable formulations.

Furthermore, advanced frameworks integrate contrastive objectives and stability losses to refine the temporal dynamics of the flow. Contrastive regularization enforces dissimilarity between positive sample flows and independent negative counterparts, ensuring the model learns distinct directional dynamics rather than collapsing to trivial solutions. Simultaneously, stability losses regularize intermediate features against latent perturbations, while progressive scheduling of regularization terms balances the trade-off between pushing the pushforward density toward the target and constraining the magnitude of transformation. Together, these mechanisms create a principled approach to managing temporal discrepancies, result-

ing in faster convergence and higher-fidelity generative performance across diverse data modalities.

2.3.2 Score Matching and Variational Bounds

Score matching provides a foundational framework for estimating the gradient of the log-density, or score function, without requiring knowledge of the intractable normalizing constant. By minimizing the Fisher divergence between the model and data distributions, this approach transforms density estimation into a tractable least-squares regression problem. In the context of continuous-time generative modeling, denoising score matching extends this principle by perturbing data with varying noise levels, effectively training a time-dependent score estimator that serves as the drift term for reverse-time stochastic differential equations.

The theoretical connection between score matching and variational inference is established through the equivalence of the score matching objective to a variational lower bound on the data log-likelihood. Specifically, optimizing the score function corresponds to maximizing an evidence lower bound (ELBO) where the approximate posterior is defined by the reverse diffusion process. This formulation reveals that the score matching loss differs from the conditional score matching loss only by a constant independent of model parameters, ensuring that gradient-based optimization yields identical updates while providing a rigorous probabilistic interpretation of the generative trajectory.

Furthermore, recent analyses demonstrate that flow matching and score matching converge to equivalent optimal vector fields under specific conditions, particularly when the source distribution is Gaussian. While flow matching directly regresses a velocity field to transport probability mass, score matching implicitly defines this transport through the gradient of the log-density. The integration of variational bounds allows for the derivation of tight error estimates on the global truncation error of numerical solvers, linking the variance of the learned drift and the local discretization error to the overall fidelity of the generated samples.

2.3.3 Consistency Distillation and One Step Generation

Consistency distillation and one-step generation frameworks address the computational inefficiency inherent in iterative sampling by learning

a direct mapping from noise to data. These methods typically employ a teacher-student paradigm where a pre-trained multi-step flow model guides a student network to approximate the solution of the underlying ordinary differential equation in a single function evaluation. By minimizing the discrepancy between the student’s output and the teacher’s trajectory endpoints, these approaches effectively collapse the generation process, theoretically enabling instantaneous sampling while attempting to preserve the high-fidelity distributional characteristics of the original model.

A critical challenge in this domain involves the trade-off between generation speed and sample quality, as aggressive distillation often leads to mode collapse or degraded visual fidelity. To mitigate these issues, advanced techniques utilize rectified flows and non-linear interpolants to straighten probability paths, thereby reducing the approximation error associated with coarse time discretization. Furthermore, simulation-free training objectives have emerged as scalable alternatives to numerical integration, allowing the model to learn consistency constraints directly from paired noise-data samples without the overhead of solving differential equations during the training phase.

Recent advancements extend these principles through sequential reflow mechanisms and segmented time divisions, which iteratively refine the vector field to enhance path straightness before final distillation. Empirical evidence suggests that while standard distillation yields significant speedups, incorporating guidance scales and optimizing the coupling between source and target distributions are essential for maintaining performance in conditional generation tasks. Consequently, modern one-step generators increasingly rely on sophisticated loss functions that enforce distributional matching rather than strict pairwise alignment, ensuring robust convergence even when the inferred generation path deviates from strict linearity.

2.4 Evaluation Metrics and Theoretical Guarantees

2.4.1 Data Efficient Distributional Metrics

Data-efficient distributional metrics address the computational bottlenecks inherent in evaluating high-dimensional generative models, particularly when sample budgets are constrained. Traditional measures like the Fréchet Inception Distance of-

ten require tens of thousands of samples to converge, prompting a shift toward metrics grounded in optimal transport theory and Stein discrepancies. By leveraging the geometric structure of the Wasserstein space, these approaches capture the cost of transporting probability mass between empirical and model distributions more effectively than moment-matching techniques, providing robust signals even with limited data availability.

Recent advancements integrate these metrics directly into the training objective of flow-based and diffusion models to enhance sample efficiency. Methods utilizing deterministic couplings or semi-discrete optimal transport plans reduce the variance associated with independent sampling from noise and data marginals. Furthermore, extensions of Stein discrepancy allow for the evaluation of unnormalized densities without requiring explicit normalizing constants, facilitating the assessment of complex manifolds. These techniques enable the construction of probability paths that align closely with iso-density surfaces, ensuring that the learned vector fields preserve structural fidelity while minimizing the number of function evaluations required for convergence.

The adoption of such metrics has demonstrated significant improvements in both training stability and generation quality across diverse benchmarks. By prioritizing distributional alignment over simple moment matching, models achieve superior performance in fidelity and diversity scores with fewer training iterations. This paradigm shift not only accelerates the development cycle but also supports the deployment of generative systems in data-scarce regimes. Consequently, data-efficient distributional metrics serve as a critical component in modern generative frameworks, bridging the gap between theoretical convergence guarantees and practical application constraints.

2.4.2 Convergence Analysis and Error Bounds

Convergence analysis in flow-based generative models primarily hinges on bounding the Global Truncation Error (GTE) introduced by numerical ODE solvers. Theoretical frameworks decompose GTE into three critical components: the Local Truncation Error (LTE) per integration step, the variance of the learned drift vector field, and the total integration interval. Rigorous bounds often rely on the Lipschitz continuity of the velocity field and the spectral norm of its Jacobian. When the drift

exhibits high variance or unbounded growth near the data manifold, the solver becomes ill-posed, leading to significant divergence between the theoretical continuous trajectory and the discrete numerical approximation.

To address numerical instability, recent works employ regularization techniques targeting the Jacobian spectral norm and utilize unbiased estimators like Hutchinson’s trace estimator for divergence calculations. While these methods mitigate memory costs during training, they introduce approximation errors that must be accounted for in convergence guarantees. Specifically, the trade-off between perturbation intervals and noise levels is crucial; setting the minimum time step too small induces unbounded negative log-likelihood behavior, whereas overly large steps degrade sampling fidelity by preventing the model from reaching the clean-data limit. Consequently, optimal convergence regimes are identified where the vector field remains sufficiently smooth for coarse-step integration without sacrificing distributional alignment.

Formal convergence guarantees are further established through Evolution Variational Inequalities (EVI) within the Wasserstein space, leveraging the λ -convexity of the Kullback-Leibler divergence. This approach mirrors proximal gradient descent analysis, demonstrating that the distance between the generated and target distributions decreases monotonically up to an error term dependent on the optimization precision. Linear convergence rates are achievable under specific assumptions regarding the source scale and the conditioning of the underlying quadratic objectives. However, existing bounds often assume idealized conditions, such as spectrally uncorrelated sources, leaving the impact of discretization errors and complex dependency structures in high-dimensional spaces as active areas for theoretical refinement.

2.4.3 Robustness to Perturbations and Pruning

Robustness to input perturbations is a critical determinant of model stability, particularly when dealing with quantized data or adversarial noise patterns. Standard practices often incorporate uniform dequantization noise during training to smooth the data manifold, yet models frequently exhibit sensitivity to structured disturbances such as checkerboard patterns or constant shifts. Recent

analyses indicate that while regularization techniques can mitigate some instability, the variance of the drift term and the stiffness of the underlying vector field significantly influence global truncation errors. Consequently, methods that enforce deviation constraints through alternating corrector steps have been proposed to steer trajectories back toward learned manifolds, effectively reducing accumulated errors caused by off-manifold drifts induced by perturbations.

The structural integrity of generative flows under pruning operations is intrinsically linked to the Lipschitz continuity of the learned mapping and the conditioning of the optimization landscape. Aggressive parameter reduction often exacerbates ill-conditioning, leading to sharp performance collapses, especially in over-guided regimes where the vector field becomes too stiff for coarse-step integration. Preconditioning strategies, such as whitening transformations, have proven essential in maintaining a stable condition number across time steps, thereby preserving reconstruction accuracy even under high compression ratios. Without such measures, the removal of parameters can disrupt the delicate balance required for straightening flows, resulting in degraded sample quality and increased susceptibility to numerical instability during inference.

Furthermore, the interplay between perturbation robustness and model sparsity reveals that simulation-free approaches, like consistency models, offer distinct advantages by learning direct mappings that bypass iterative solver accumulation errors. However, these models remain vulnerable to specific failure modes when the coupling between noise and data distributions is not optimally aligned. Advanced reparameterization techniques that restrict transformations to additive shifts help eliminate trivial zero-cost solutions while retaining volume-preserving properties essential for robust pruning. Ultimately, achieving resilience against both external perturbations and internal structural reductions requires a holistic design that integrates stable numerical solvers, condition-aware parameterizations, and rigorous bounds on drift variance to ensure reliable performance across diverse operational constraints.

3 Controllable Image Generation via Diffusion and Flow Models

3.1 Architectural Mechanisms for Spatial Control

3.1.1 Attention Based Feature Injection and Modulation

Attention-based feature injection mechanisms have emerged as a pivotal strategy for integrating auxiliary conditions into diffusion backbones without compromising pretrained knowledge. Unlike dense concatenation approaches, methods such as GLIGEN introduce trainable gated self-attention layers within transformer blocks to process discrete grounding tokens, such as bounding boxes paired with semantic embeddings [2]. These gates are typically initialized to zero, ensuring that the model retains its original generative distribution during early training stages while gradually learning to modulate latent representations based on explicit spatial or semantic constraints.

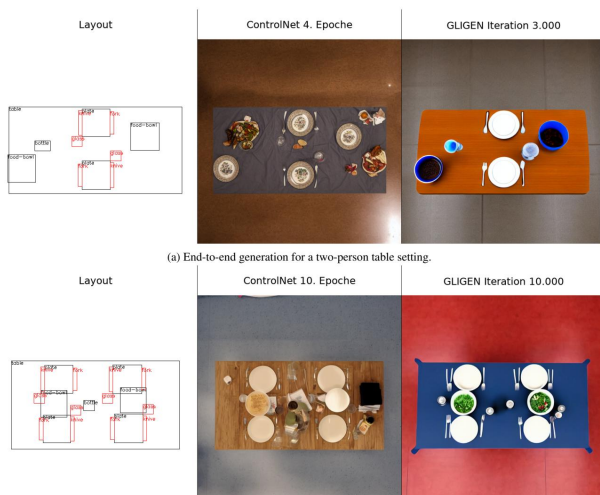


Figure 3: Chart from 'A Two-Stage System for Layout-Controlled Image Generation Using Large Language Models and Diffusion Models' (arXiv:2511.06888)

Beyond discrete token injection, recent advancements focus on modulating cross-attention maps to achieve fine-grained structural control and consistent image editing. Techniques like Prompt-to-Prompt and MasaCtrl manipulate attention weights directly to preserve spatial layouts or transfer subject identities across generations by aligning query-key interactions between source and target domains. This modulation often involves replacing standard attention inputs with fused features derived from reference images or control signals, allowing the model to attend to

specific regions defined by masks or edge maps while maintaining high-fidelity texture synthesis in unconstrained areas.

Furthermore, multi-scale conditional injection via Attention Feature Fusion modules enables the seamless integration of control features at various resolutions within the U-Net architecture. By reshaping extracted condition vectors to match current noise dimensions and fusing them through attention mechanisms, these approaches facilitate dynamic weighting of guidance signals throughout the denoising trajectory. This flexibility allows for precise balancing between adherence to input constraints, such as sketches or depth maps, and the preservation of semantic diversity, ultimately enhancing controllability in complex generation tasks like terrain synthesis and multi-subject composition.

3.1.2 Adapter Networks and Multi Condition Fusion

Adapter networks have emerged as a parameter-efficient paradigm for injecting conditional signals into pretrained diffusion backbones without altering the original weights. By freezing the core generative model and training lightweight auxiliary modules, these architectures map diverse control inputs—such as edge maps, depth estimates, or semantic layouts—into the latent feature space. The prevailing mechanism involves computing residual modifications to intermediate layer activations, which are then element-wise added to the host network’s features. This design preserves the fidelity of the underlying quality manifold while enabling precise structural guidance, effectively decoupling the learning of control semantics from the generative prior.

To address the complexity of simultaneous multi-condition generation, recent advancements focus on sophisticated fusion strategies within these adapter frameworks. Rather than relying on naive concatenation or sequential application, modern approaches employ joint training designs that utilize unified encoders to embed heterogeneous conditions into a shared representation space. Techniques such as orthogonal adaptation and weight fusion are increasingly adopted to mitigate interference between conflicting control signals, ensuring that the integration of multiple constraints—like pose, lighting, and geometry—remains coherent. These methods often leverage attention-based mechanisms or hypernetworks to

dynamically modulate the influence of each condition based on contextual relevance.

Despite their efficacy, current adapter-based fusion faces challenges regarding scalability and computational overhead, particularly as the number of control modalities increases. While all-in-one adapters aim to handle arbitrary condition types within a single network, they often struggle to maintain performance parity with modality-specific counterparts due to capacity constraints. Furthermore, the fixed-parameter nature of the backbone can limit the extent of controllability when intermediate features are heavily perturbed. Consequently, ongoing research explores geometry-aware constraints and consistency-based feedback loops to enhance the robustness of multi-condition fusion, striving to balance diverse control capabilities with high-fidelity image synthesis.

3.1.3 Layer Decomposition and Occlusion Handling

Layer decomposition strategies address the complexity of scene synthesis by disentangling foreground objects from backgrounds and intermediate layers, enabling fine-grained compositional control. By factorizing the generation process into spatial regions or distinct semantic layers, these methods leverage the complementary strengths of large language models for spatial reasoning and diffusion architectures for visual fidelity. This decomposition allows for the independent manipulation of specific scene elements, such as assigning unique textual prompts to spatial sub-regions or scaling factors, effectively recovering specialized techniques like MultiDiffusion as particular instances of a broader factorized framework.

Occlusion handling remains a critical challenge within this paradigm, particularly when managing high object density or strict compositional constraints where standard segmentation masks may prove inaccurate. Traditional approaches often rely on predicting intermediate geometric representations, such as ambient occlusion maps, surface normals, and depth estimates, to resolve visibility conflicts before rendering final shadows or reflections. However, these multi-stage pipelines can be computationally prohibitive and sensitive to view-angle variations, prompting a shift towards end-to-end solutions that integrate structural priors directly into the denoising trajectory to maintain consistency without explicit intermediate ren-

dering steps.

To mitigate artifacts arising from imperfect masks or latent space compression, advanced sampling techniques enforce layout adherence early in the diffusion process, ensuring that global silhouettes are established before fine-grained details are synthesized. Mechanisms such as locality-aware mask adaptation and direct sketch guidance integration help preserve structural consistency between conditional inputs and generated latents, reducing distortions in overlapping regions. Despite these advances, current methods still struggle with extremely fine details and complex inter-layer interactions, suggesting that future iterations must focus on mask-free approaches and iterative refinement strategies to robustly handle occlusion in highly cluttered scenes.

3.2 Semantic and Attribute Guided Synthesis

3.2.1 Text Driven Layout and Object Grounding

Text-driven layout and object grounding have emerged as pivotal mechanisms for enhancing spatial controllability in generative models. Early approaches leveraged Large Language Models to translate natural language prompts into structured layout representations, such as CSS-like formats or bounding box coordinates, effectively bridging the gap between semantic intent and spatial arrangement [2]. By utilizing in-context learning, these systems enable training-free layout planning that decomposes complex multi-object scenes into phrase-by-phrase spatial specifications, allowing for implicit reasoning about object relationships without extensive fine-tuning [2].

To operationalize these layouts within diffusion frameworks, recent methodologies integrate grounded language directly into the generation process through specialized architectural modifications. Techniques such as gated attention mechanisms inject bounding box and text embedding pairs into trainable layers, enabling open-set grounding where objects are defined by free-form descriptions rather than fixed class labels. This integration allows the model to simultaneously process global textual context and local spatial constraints, ensuring that generated content adheres strictly to specified locations while maintaining semantic consistency with the overall scene description.

Advanced implementations further refine this

control by incorporating visual reference banks and optimizing supervision strategies to handle occlusion and object interaction. By shifting from standard epsilon-supervision to direct data prediction, these models significantly reduce memory requirements while improving convergence on dense grounding tasks. The fusion of layout embeddings with multimodal attention modules facilitates a deeper understanding of scene geometry, enabling the synthesis of high-fidelity images where object placement, scale, and inter-object relationships are precisely governed by the input text and spatial priors.

3.2.2 Fine Grained Attribute Editing and Personalization

Fine-grained attribute editing has evolved from coarse prompt engineering to precise, instance-level manipulation leveraging disentangled representations. Contemporary approaches utilize structured prompt systems and attention mechanisms to isolate specific semantic attributes, such as color or texture, while preserving the integrity of unrelated image regions. By integrating spatial guidance like bounding boxes or segmentation masks, these methods enable users to modify individual object instances within a complex scene without inducing global stylistic shifts or hallucinations, effectively balancing default-mode diversity with explicit controllability.

Personalization techniques address the challenge of adapting pre-trained diffusion models to novel concepts using limited data through parameter-efficient fine-tuning strategies. Methods such as Low-Rank Adaptation and textual inversion inject new semantic information into the model’s weight space or learn specialized token embeddings, allowing for the faithful reproduction of specific subjects or styles. However, these approaches often face a trade-off between plasticity and stability; while they excel at capturing unique identity features, frequently adding or recombining personalized concepts can lead to catastrophic forgetting or interference with the model’s original generative distribution.

Recent advancements aim to resolve these limitations by decoupling attribute control from identity preservation through modular architectures and regularization losses. Instead of permanently altering base weights, emerging frameworks employ plug-and-play modules that encode normalized aesthetic values or utilize test-time adapta-

tion to dynamically adjust generation parameters. This shift facilitates continuous modulation of fine-grained details, such as lighting and material properties, ensuring high-fidelity personalization that maintains compositional accuracy and reduces detail distortion without requiring extensive retraining on specialized datasets.

3.2.3 Style Transfer and Artistic Concept Alignment

Style transfer in diffusion models has evolved from simple feature statistics matching to sophisticated attention-based mechanisms that preserve semantic integrity while imposing artistic aesthetics. Recent approaches leverage shared attention layers to align the style of generated images with reference inputs without requiring extensive fine-tuning. By normalizing queries and keys through Adaptive Instance Normalization within the cross-attention modules, these methods effectively decouple content structure from stylistic texture, allowing for the seamless integration of complex artistic concepts into diverse semantic layouts.

A critical challenge in this domain is the "illusion of variety," where generative trajectories collapse into high-probability modes, limiting the exploration of a concept’s true semantic space [3]. To counteract this, advanced frameworks introduce orthogonal control mechanisms that force interacting generation paths to diverge, thereby covering a wider range of stylistic interpretations. This ensures that while the model adheres to specific artistic constraints, it maintains sufficient stochasticity to produce distinct outputs that reflect the full breadth of the target style rather than converging on a single, repetitive aesthetic prototype.

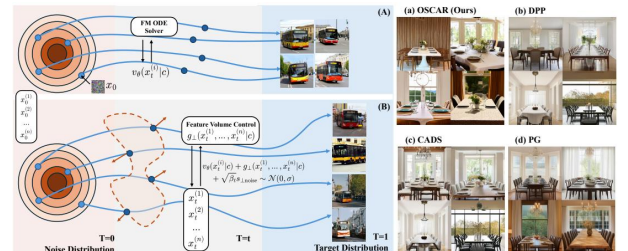


Figure 4: Chart from 'OSCAR: ORTHOGONAL STOCHASTIC CONTROL FOR ALIGNMENT-RESPECTING DIVERSITY IN FLOW MATCHING' (arXiv:2510.09060)

Furthermore, achieving fine-grained alignment often involves quantifying abstract aesthetic at-

tributes using multimodal representations from pretrained vision-language models. Instead of relying solely on ambiguous textual prompts or costly human preference data, modern techniques utilize similarity-based estimation to guide the generation process toward desired stylistic intensities. This plug-and-play capability allows for continuous control over aesthetic properties, balancing realism and artistic flair while preventing the hallucination of objects that do not align with the underlying structural conditions or layout constraints.

3.3 Training Free and Inference Time Control

3.3.1 Latent Space Manipulation and Noise Initialization

Latent space manipulation fundamentally shifts the diffusion process from high-dimensional pixel spaces to compressed representations learned by variational autoencoders, significantly enhancing computational efficiency while preserving semantic fidelity [4]. In this regime, initial noise samples are drawn from a standard Gaussian distribution and mapped into the latent manifold, where the denoising trajectory is governed by a velocity field model conditioned on auxiliary inputs such as text embeddings or spatial masks. This compression allows for precise control over generative outcomes by operating directly on the feature embeddings rather than raw pixels, enabling the integration of complex conditioning signals through channel concatenation or cross-attention mechanisms within the underlying U-Net architecture.

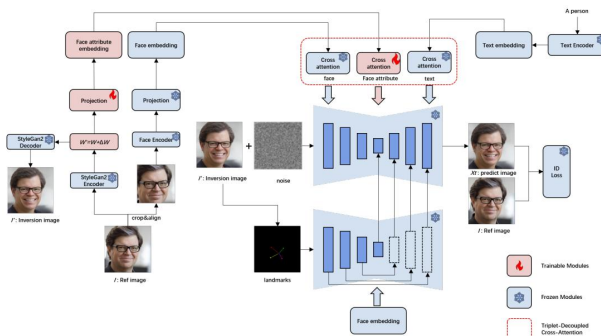


Figure 5: Chart from 'PSTF-AttControl: Per-Subject-Tuning-Free Personalized Image Generation with Controllable Face Attributes' (arXiv:2510.25084)

Advanced initialization strategies extend beyond simple Gaussian sampling to actively shape the generative trajectory through latent space op-

timization. Techniques such as second-order Heun extrapolation predict local endpoints during early time steps, allowing for the maximization of feature-space volume defined via the log-determinant of centered Gram matrices. By pulling back the gradient of this volume potential to the latent space, the system encourages diverse exploration and prevents mode collapse. Furthermore, scaling the magnitude of the initial latent noise serves as a critical hyperparameter; reducing the noise scale monotonically decreases fine-grained detail in the final output while maintaining global semantic structure, offering a direct mechanism for controlling image complexity without re-training.

The efficacy of these manipulation techniques relies heavily on the structural properties of the latent representation, particularly when paired with robust feature encoders that concentrate samples associated with identical conditioning signals. Methods leveraging structured latent spaces, such as those derived from representation autoencoders, reduce multimodality and facilitate more stable convergence during the reverse diffusion process. By aligning the score functions within these compressed domains and employing guided endpoint means under Gaussian posterior approximations, models achieve superior alignment between the generated distribution and the target conditional density. This synergy between noise initialization protocols and latent geometry ensures that the generative process remains both computationally tractable and semantically coherent across diverse synthesis tasks.

3.3.2 Gradient Based Guidance and Reward Optimization

Gradient-based guidance represents a paradigm shift from passive conditioning to active optimization, directly manipulating the sampling trajectory via diversity objectives or reward signals. Early approaches like Particle Guidance enforced repulsion directly in the latent space to encourage exploration; however, this often conflicted with the model's intrinsic quality-generating flow, leading to mode collapse or artifact generation. To mitigate these adversarial effects, contemporary methods such as DiverseFlow elevate the guidance objective, decoupling diversity constraints from the primary denoising process to maintain fidelity while expanding the support of the generated distribution.

The mathematical foundation of this approach typically involves augmenting the standard score function with an auxiliary gradient term derived from a differentiable reward model or classifier. While classifier-guided mechanisms theoretically offer precise control by scaling the log-likelihood gradient, they frequently struggle with robustness against high-noise regimes and may introduce adversarial artifacts when irrelevant features dominate the prediction. Consequently, the field has increasingly adopted classifier-free guidance and direct reward optimization techniques, which approximate the conditional score without requiring an external, separately trained classifier, thereby stabilizing the backward diffusion process and improving alignment with complex semantic prompts.

Recent advancements further refine this mechanism through time-scheduled stochastic noise and controllable softness parameters, allowing for dynamic adjustment of the guidance scale throughout the inference steps. By integrating these gradients into the update rule, models can effectively navigate the trade-off between strict adherence to structural constraints—such as layout bounding boxes or depth maps—and the preservation of perceptual quality. This optimization framework enables fine-grained control over generation attributes, including shadow direction and texture softness, ensuring that the final samples satisfy both global semantic requirements and local geometric constraints without necessitating extensive task-specific fine-tuning.

3.3.3 Attention Map Refinement and Self Guidance

Attention map refinement has emerged as a pivotal training-free strategy for enhancing semantic alignment in diffusion models without incurring additional computational costs during inference. By explicitly modulating cross-attention weights, these methods rectify the tendency of standard sampling to overlook specific prompt tokens, thereby ensuring that generated content adheres strictly to textual descriptions. Techniques often involve injecting learnable tokens or applying mutual self-attention constraints to force query tokens to attend to correct reference regions, effectively resolving issues of object omission or attribute binding errors common in baseline architectures.

Complementing spatial refinement, self-

guidance mechanisms address the limitations of strong Classifier-Free Guidance, which often narrows the explored manifold and reduces semantic diversity despite improving prompt fidelity. Instead of relying solely on external conditions, self-guidance leverages the model’s own intermediate predictions to steer the denoising trajectory, balancing realism with structural consistency. This approach frequently employs score alignment or prior guidance to adjust latent codes, ensuring that the generation process maintains high-frequency details while avoiding the mode collapse associated with over-weighted conditional scores.

The integration of refined attention maps with self-guidance principles enables precise control over both global layout and local attributes. Recent advancements combine these strategies by enforcing orthogonality between guidance vectors and the base flow, allowing for controllable noise injection that boosts diversity without compromising structural integrity. Such hybrid frameworks facilitate attribute-level manipulation and complex scene composition, demonstrating that manipulating internal attention dynamics offers a more efficient path to controllability than training extensive external adapter networks or relying on naive multi-sampling remedies.

3.4 Domain Specific Applications and Constraints

3.4.1 Medical Imaging and Anatomical Fidelity

In the domain of medical imaging, achieving high anatomical fidelity presents unique challenges distinct from natural image synthesis, primarily due to the critical need for structural correctness over aesthetic appeal. Unlike color or style attributes which admit straightforward perceptual metrics, verifying whether a generated silhouette matches a target organ shape or if limbs are correctly oriented lacks a simple scalar proxy. Recent studies indicate that even advanced vision-language models struggle to reliably identify relative positions and geometric constraints within medical scans, often failing to distinguish left from right or misaligning subtle anatomical features. This limitation underscores the difficulty of applying standard reward- and gradient-based approaches to tasks requiring precise geometric control.

To address these deficiencies, contemporary dif-

fusion frameworks increasingly integrate explicit structural conditioning mechanisms, such as skeletal guides, normal maps, and depth representations, to enforce spatial consistency. Architectures adapted from segmentation networks, particularly U-Net variants, are frequently employed to preserve fine-grained details essential for diagnostic accuracy. However, standard denoising processes often suffer from low learning signals during early timesteps, which are crucial for determining global composition. Consequently, novel loss functions and guidance strategies have been developed to suppress distortions in complex structures like hands, faces, or specific organ boundaries, ensuring that the generated output maintains topological integrity throughout the synthesis process.

Despite these advancements, balancing image quality with strict adherence to conditional inputs remains a persistent hurdle. Ablation studies reveal that removing specific fidelity safeguards can severely harm the equilibrium between generation diversity and anatomical precision, leading to hallucinated structures or visible artifacts. While methods utilizing multi-distributed diffusion or transformer-based backbones show promise in rivaling large-scale capabilities, they must explicitly disentangle fine-grained anatomical factors to avoid the blurry or cartoon-like tendencies associated with maximizing likelihood in high-density regions. Future directions necessitate robust evaluation metrics that go beyond pixel-level similarity to assess semantic plausibility and layout adherence, ensuring generated medical images are both photorealistic and clinically viable.

3.4.2 Remote Sensing and Geospatial Conditioning

Remote sensing and geospatial conditioning leverage multi-modal spatial priors to guide diffusion models in generating high-fidelity satellite and aerial imagery. Unlike standard semantic controls, these approaches integrate complex geospatial data such as land cover maps, digital elevation models, and hyperspectral signatures directly into the denoising process. By aligning the latent space with specific geographic coordinates and topological features, models can synthesize realistic terrain patterns that adhere to strict physical constraints, ensuring that generated outputs maintain consistency with real-world climatic and geological distributions.

The technical implementation typically in-

volves adapter networks or modified cross-attention mechanisms designed to process high-dimensional geospatial inputs alongside textual prompts. These architectures often employ spatial decomposition strategies to handle the vast scale of panoramic satellite data, allowing for region-specific conditioning where local features like water bodies or urban structures are governed by distinct mask-guided priors. Furthermore, advanced methods utilize factorized diffusion to manage the consensus of noise estimates across overlapping crops, effectively resolving boundary artifacts and ensuring seamless transitions in large-scale map generation without compromising structural integrity.

Evaluation in this domain extends beyond perceptual quality to include rigorous distributional coverage metrics tailored to geospatial fidelity. Techniques such as Improved Precision-Recall curves quantify the model's ability to capture the breadth of the true data manifold while preserving low-level details critical for downstream analysis. Recent advancements demonstrate that incorporating explicit geometric factors and multi-level denoising synthesizers significantly reduces ambiguity in ill-conditioned scenarios, enabling precise control over diverse modes such as seasonal variations or specific land-use changes. This capability broadens the applicability of generative models for simulation, planning, and environmental monitoring tasks.

3.4.3 Human Pose and Identity Consistency

Achieving simultaneous consistency in human pose and identity remains a critical bottleneck in controllable person image generation. While modern diffusion frameworks effectively adhere to spatial constraints via pose keypoints, they frequently suffer from an "illusion of variety," where outputs collapse into high-probability modes that fail to explore the full semantic space of the target concept. This mode collapse often manifests as a disconnect between the generated figure and its environment; although the skeletal structure aligns with conditioning signals, the subject appears detached or "sticker-like," lacking coherent integration with background lighting and perspective.

The challenge is exacerbated by the inherent ambiguity in mapping abstract aesthetic attributes to precise anatomical structures without compromising identity fidelity. Existing methods utilizing gradient-based optimization or reward models

struggle to define scalar proxies for structural correctness, often resulting in distorted textures or misplaced limbs when attempting fine-grained manipulation. Furthermore, preserving unique facial features during aggressive pose transfer requires balancing latent code scaling with identity embeddings, a trade-off that frequently leads to the degradation of personal characteristics or the introduction of artifacts in complex multi-person scenarios.

To mitigate these issues, recent approaches have shifted towards plug-and-play frameworks that decouple appearance control from geometric constraints. By leveraging dense intrinsic appearance flows and fixed identity embeddings, these systems aim to inject structural priors without necessitating extensive retraining or additional model evaluations. However, while such techniques improve local texture preservation and reduce distortion, broader quantitative evaluation indicates that maintaining global scene coherence and robust identity consistency across diverse, unconstrained poses remains an open research direction requiring more sophisticated disentanglement of style and content representations.

4 Flow-Based Generative Modeling for Video, Speech, and Cross-Modal Synthesis

4.1 Temporal Coherence and Motion Control

4.1.1 Optical Flow Integration and Camera Trajectory

Optical flow integration serves as a critical mechanism for embedding explicit motion priors into generative trajectories, effectively bridging the gap between static image synthesis and dynamic video generation. By extracting dense optical flow fields from source videos using estimators like Unimatch, models can condition the velocity field of flow matching or normalizing flows on these spatial displacements. This approach allows the generative process to sample trajectories that strictly adhere to observed pixel-wise movements, ensuring temporal consistency and enabling the synthesis of complex motion dynamics without relying solely on implicit latent space interpolation.

The formulation of camera trajectory within this framework often leverages the invertible properties of normalizing flows to map 3D camera parameters directly to 2D motion patterns. By defining camera motion through sequences of extrinsic

and intrinsic parameters, the system can generate fixed optical flow patterns corresponding to specific maneuvers such as panning, zooming, or rotation. These predefined flow vectors act as steering conditions within the vector field, guiding the integration of the ordinary differential equation from noise to data along a path that reflects the desired cinematic movement, thereby offering precise control over the generated video's perspective and depth.

Furthermore, recent advancements in rectified flow and optimal transport aim to linearize these integration paths, minimizing the number of inference steps required while maintaining high fidelity [1]. However, challenges such as "trajectory lock-in" remain, where the initial velocity field excessively constrains the global tangent direction, potentially causing premature collapse of the latent volume. To mitigate this, techniques involving gradient skipping and behavior-corrected transition ratios are employed to regularize the predicted velocity field, ensuring that the camera trajectory remains flexible enough to accommodate open-domain variations while preserving the structural integrity of the intended motion.

4.1.2 Autoregressive Latent Dynamics for Video

Autoregressive latent dynamics for video generation operate by compressing high-dimensional spatiotemporal data into a compact, continuous latent space using pre-trained variational autoencoders. This dimensionality reduction preserves perceptual quality while enabling efficient modeling of complex motion patterns that are computationally prohibitive in pixel space. By treating each latent vector as a discrete token or continuous state sampled from a structured distribution, these methods decouple the representation learning phase from the temporal dynamics modeling, allowing for the seamless integration of powerful image priors into video synthesis frameworks.

Within this compressed domain, autoregressive models predict future latent states sequentially based on historical context, effectively capturing long-term temporal dependencies through recurrent neural networks or transformer architectures. Unlike diffusion-based approaches that denoise entire sequences simultaneously, autoregressive latent dynamics facilitate frame-by-frame generation, offering inherent support for variable-length video synthesis and real-time inference capabilities.

ties. The sequential nature of this process allows for precise control over motion trajectories by manipulating latent variables at each time step, although it introduces challenges regarding error accumulation and the maintenance of global consistency over extended durations.

Recent advancements have focused on enhancing the expressiveness of these latent transitions by incorporating invertible transformations and flow-based corrections to better approximate the true posterior distribution of motion dynamics. Techniques such as masked autoregressive flows impose strict invertibility on the factorization process, bridging the gap between explicit density estimation and sequential prediction. Furthermore, integrating motion-conditioned priors and rectified flow objectives within the latent space has significantly improved the stability of long-horrollouts, mitigating the unnatural distortions often associated with traditional warping operations while maintaining the computational efficiency required for high-resolution video generation.

4.1.3 Motion Field Estimation and Animation

Motion field estimation in modern generative frameworks increasingly leverages flow matching to establish deterministic trajectories between latent distributions. By parameterizing a time-dependent velocity field, these methods transport a simple base distribution to complex data manifolds along prescribed paths, often utilizing optical flow estimators to extract dense displacement fields from video sequences. This approach facilitates the construction of linear or rectified trajectories that significantly reduce sampling steps compared to traditional diffusion processes, enabling efficient interpolation between distinct motion states while preserving temporal coherence.

In the context of animation and synthesis, the learned velocity field governs an ordinary differential equation that drives the transformation of static conditions into dynamic sequences. Techniques such as Motion Flow Matching attempt to generate poses from textual inputs by modeling the vector field in latent space, though challenges remain in maintaining visual appearance and adhering to strict pose constraints without producing mannequin-like artifacts. Advanced architectures address this by incorporating conditional vectors and alternating autoregressive orderings, allowing for precise control over motion attributes while en-

suring the generated frames align with specified semantic or spatial guidance.

To enhance fidelity and control, recent developments integrate multi-scale decompositions, such as wavelet or Fourier transforms, to organize scaling bands within the velocity field learning process. This allows the model to capture both global motion dynamics and fine-grained local details effectively. Furthermore, adaptive integration solvers and specialized training trajectories, including sine-based interpolations, are employed to optimize the curvature of the flow path. These refinements ensure stable convergence and high-quality animation synthesis, bridging the gap between abstract motion representations and realistic, temporally consistent video generation.

4.2 Audio and Speech Synthesis via Flows

4.2.1 Text to Speech and Prosody Control

Recent advancements in text-to-speech (TTS) have shifted towards flow matching and diffusion-based architectures to achieve high-fidelity, multilingual speech generation at scale [5]. Models such as Voicebox and Audiobox leverage non-autoregressive flow matching to synthesize audio directly from text prompts, enabling efficient inference and robust handling of diverse linguistic inputs. These systems utilize large-scale pre-training on universal speech corpora, allowing them to generalize across languages and speakers without requiring language-specific fine-tuning, thereby establishing a new baseline for controllable audio synthesis.

$$M_y = \mathcal{E}(y),$$

Figure 6: Chart from 'FlowSE: Efficient and High-Quality Speech Enhancement via Flow Matching' (arXiv:2505.19476)

Prosody control within these frameworks is increasingly managed through classifier-free guidance (CFG) and explicit conditioning mechanisms. By adjusting guidance weights during the denoising or flow integration process, models can modulate emotional tone, speaking rate, and intonation without compromising audio quality. Unlike traditional methods that rely on separate pitch or duration predictors, modern approaches embed

prosodic constraints directly into the latent space or use natural language descriptions as soft controls. This flexibility allows for a wider range of expressive outputs while maintaining stability across varying guidance scales, addressing the common trade-off between controllability and naturalness.

Furthermore, the integration of orthogonal semantic signals has been proposed to mitigate mode collapse and enhance the distinctness of generated prosodic styles. By injecting disentangled control vectors into the generation trajectory, systems can precisely manipulate specific acoustic attributes while preserving global coherence. This strategy, combined with task-agnostic architectures that avoid specialized decoder heads, facilitates a unified framework for both speech and general audio generation. Consequently, current research emphasizes the development of adaptive modulation parameters and spatially-aware conditioning to refine the granularity of prosodic control in real-time applications.

4.2.2 Speech Enhancement and Intelligibility Restoration

Recent advancements in speech enhancement have increasingly leveraged generative modeling to address the limitations of traditional linear estimators, particularly in low signal-to-noise ratio regimes [5]. While classical methods like Least Squares suffer from severe degradation under significant noise, modern approaches utilize diffusion probabilistic models and flow-based architectures to perform robust denoising [5]. These techniques frame enhancement as a conditional generation task, where iterative posterior sampling or normalizing flows transform noisy inputs into clean speech distributions, effectively compensating for signal ambiguity where pilot observations are sparse.

A critical challenge in this domain involves balancing spectral fidelity with computational efficiency during the reverse denoising process. Techniques such as classifier-free guidance and latent diffusion have been adapted to operate in compressed latent spaces, significantly accelerating inference while maintaining high-quality reconstruction. However, improper integration of noise schedules or insufficient denoising steps can introduce artifacts like over-smoothing or reduced contrast in the spectro-temporal domain. Consequently, recent works emphasize the importance

of precise noise scale selection and multi-level feature injection within U-Net backbones to preserve fine-grained acoustic details essential for natural perception.

Beyond mere noise suppression, contemporary research prioritizes intelligibility restoration as a primary objective for real-world deployment. Evaluation metrics have shifted towards word error rates derived from automatic speech recognition systems, ensuring that enhanced audio remains comprehensible to both humans and machines. Methods incorporating semantic guidance or multimodal context demonstrate superior performance in recovering dysarthric or heavily corrupted speech, often outperforming baselines even when reference speakers differ. This adaptability suggests that future systems must integrate environmental semantics to guide the generative process, ensuring robustness across diverse acoustic conditions without requiring extensive speaker-specific fine-tuning.

4.2.3 Singing Voice and Multilingual Generation

Recent advancements in singing voice and multilingual speech generation have shifted from discrete token modeling to continuous flow-based architectures, enabling high-fidelity audio synthesis at scale. Systems like Voicebox employ non-autoregressive flow matching to achieve text-guided generation across multiple languages without requiring paired text-audio data for every target tongue [5]. This methodology leverages large-scale self-supervised pre-training to learn a universal speech representation, allowing the model to synthesize natural prosody and timbre while maintaining linguistic accuracy in zero-shot cross-lingual scenarios.

The integration of discriminator cooperative unlikelihood prompt-tuning further refines controllability within these generative frameworks. By incorporating human preference signals and adversarial discrimination during the fine-tuning phase, models can better adhere to specific stylistic constraints, such as emotional tone or singer identity, while suppressing hallucinated artifacts. This approach addresses the limitations of pure maximum likelihood estimation, ensuring that the generated singing voice preserves both the semantic content of the lyrics and the nuanced acoustic characteristics required for realistic musical performance.

Moreover, unified audio generation frameworks

now support complex conditioning mechanisms, including natural language prompts that dictate genre, instrumentation, and vocal texture simultaneously. These systems utilize latent space interpolation to facilitate smooth transitions between different singing styles or languages within a single utterance. The scalability of these transformer-based models, combined with efficient sampling strategies like straight-through estimation, significantly reduces inference latency, making real-time multilingual singing synthesis feasible for interactive applications and large-scale content creation pipelines.

4.3 Cross Modal and Multimodal Generation

4.3.1 Molecule to Image and Structure Alignment

The alignment of molecular structures with visual representations constitutes a critical challenge in generative modeling, requiring the precise mapping of 3D atomic coordinates to 2D image manifolds. This process typically involves encoding the source molecule, defined by atom positions $x \in \mathbb{R}^{N \times 3}$ and properties h , into a latent space that preserves both geometric fidelity and semantic identity. By leveraging pretrained encoders such as DINOv2 for global feature extraction and variational autoencoders for local pose embedding, frameworks establish a robust correspondence between the physical structure and its target image projection. This dual-encoding strategy ensures that the generated output adheres strictly to the target pose while maintaining the intrinsic appearance characteristics of the source molecular data.

To achieve effective alignment, recent methodologies employ flow matching and normalizing flows to model the joint distribution of atomic configurations and image pixels. These approaches facilitate a smooth trajectory between the source domain and the target distribution, avoiding the abrupt transitions often observed in standard diffusion inversion techniques. By interpolating between learned embeddings or utilizing rectified flow paths, the model can generate physically valid atomic positions that correspond accurately to specific visual features. This capability is particularly evident in tasks requiring structured transformations, where the model must maintain class separability and structural consistency even under significant perturbations or varying noise levels.

Furthermore, the integration of multiscale fu-

sion mechanisms enhances the resolution and detail of the synthesized images, ensuring that fine-grained structural nuances are preserved during the generation process. Techniques such as 3D-RoPE alignment and center-cropping strategies stabilize optimization by matching token counts across different aspect ratios, thereby maximizing computational efficiency without sacrificing geometric accuracy. The resulting frameworks demonstrate superior performance in producing high-resolution outputs that faithfully reflect the underlying molecular geometry, effectively bridging the gap between abstract chemical data and concrete visual representations through rigorous structure-to-image alignment protocols.

4.3.2 Layout to Image and Document Synthesis

Layout-to-image synthesis has evolved from adversarial frameworks to sophisticated flow-based architectures that map semantic layouts to photo-realistic outputs. Early methodologies predominantly utilized Generative Adversarial Networks with spatially-adaptive normalization to enforce structural consistency, yet often struggled with mode collapse and limited diversity. Recent advancements have integrated Normalizing Flows and flow matching techniques to leverage invertible properties, enabling precise control over the generation process while maintaining high-fidelity details. These models effectively address the inverse problem of semantic segmentation by learning bijective mappings between latent distributions and image spaces conditioned on layout constraints.

The transition toward document and complex scene synthesis necessitates robust handling of long-range dependencies and multi-object relationships. Contemporary approaches employ attention mechanisms within flow-based models to capture global context, ensuring coherent text rendering and object placement across high-resolution canvases. By utilizing conditional aggregation structures, these frameworks dynamically adjust feature representations based on input semantic maps, significantly improving performance in tasks such as image inpainting and instruction-based editing. This flexibility allows for the seamless integration of diverse modalities, including depth maps and trajectory controls, facilitating fine-grained manipulation of both spatial and temporal elements in generated content.

Furthermore, the efficiency of modern layout-to-image systems has been enhanced through scalable architectures that reduce computational overhead without compromising quality. Techniques such as one-step generation and latent space optimization have demonstrated competitive performance against traditional diffusion models, particularly under high guidance weights. The ability to project real images into latent spaces for subsequent editing underscores the versatility of these methods in practical applications ranging from digital content creation to medical imaging. As research progresses, the unification of linear inverse problems and style transfer within a single flow-based framework promises to further streamline the synthesis of complex, structured documents and scenes.

4.3.3 Vision Language Action and Robot Control

Recent advancements in vision-language-action (VLA) frameworks have shifted robot control from discrete command execution to continuous flow-based generation within unified latent spaces. By treating action trajectories as executable variables embedded directly into the generative process, these models enable role-structured planning where visual perception and linguistic instructions jointly condition motor outputs. This paradigm allows for the synthesis of complex manipulation sequences without relying on intermediate reasoning tokens, effectively bridging the gap between high-level semantic intent and low-level kinematic constraints through a single, cohesive architecture.

A critical innovation in this domain involves the decoupling of control attributes into a normalized prior space while maintaining the pre-trained language model in its original latent configuration. Traditional interpolation strategies become tractable within this structured prior, facilitating explainable adjustments to motion dynamics such as timing, velocity, and spatial positioning. Instead of costly parameter fine-tuning or reinforcement learning updates, contemporary approaches leverage fixed backbone models, injecting control signals via classifier-free guidance or velocity modulation to steer the generation trajectory. This ensures that the system retains the broad generalization capabilities of large-scale pre-training while achieving precise, real-time adaptability to specific robotic tasks.

Furthermore, the integration of invertible atten-

tion mechanisms and non-local flow matching enhances the system’s ability to model long-range temporal dependencies and multi-object interactions essential for dexterous manipulation. By establishing a bi-directional mapping between semantic layouts and action flows, these systems can simultaneously optimize for camera motion, object trajectories, and end-effector poses. The resulting frameworks support tightly coupled vision-language applications where feedback from the physical environment is rapidly propagated back into the latent space, enabling robust closed-loop control that maintains stability and coherence even under dynamic environmental perturbations.

4.4 Latent Space Representations and Efficiency

4.4.1 Hierarchical Priors and Multi Scale Flows

Hierarchical priors address the expressiveness limitations of standard Gaussian bases in normalizing flows by introducing layered latent variables that capture complex, multi-modal data distributions. Unlike traditional single-level formulations, these architectures decompose the generative process into a cascade of conditional distributions, allowing higher-level latents to govern global semantic structures while lower levels refine local details. This stratification enables the model to learn richer representations without sacrificing the tractable likelihood estimation inherent to flow-based methods, effectively bridging the gap between variational autoencoders and invertible transformations.

Multi-scale flows extend this capability by operating across distinct spatial resolutions, typically employing coupling layers with split operations to progressively factor out information at each scale. By processing features from coarse to fine granularities, these architectures efficiently model long-range dependencies and high-frequency textures that single-scale networks often miss. The integration of invertible convolutions and autoregressive components within this hierarchy ensures that dimensionality is preserved while facilitating efficient inference and sampling, making them particularly suitable for high-resolution image synthesis and dense prediction tasks.

Recent advancements unify hierarchical priors with multi-scale dynamics through continuous normalizing flows and flow matching techniques, en-

abling the modeling of arbitrary probability paths in latent space. These approaches leverage ordinary differential equations to define smooth trajectories between simple base distributions and complex data manifolds, often utilizing numerical solvers for inference. By combining data-dependent deterministic priors with multi-scale architectural designs, modern frameworks achieve superior sample quality and computational efficiency, effectively mitigating the stability issues associated with deep discrete flow stacks while maintaining exact likelihood computation.

4.4.2 Energy Based Models in Latent Spaces

Energy-based models (EBMs) in latent spaces leverage the compressed, semantically rich representations provided by variational autoencoders to define probability distributions via unnormalized density functions. By operating within these lower-dimensional manifolds, EBMs avoid the computational intractability associated with high-dimensional pixel spaces, focusing instead on modeling complex attribute distributions through an energy function that assigns low values to realistic latent codes. This approach facilitates precise control over generative trajectories, as the energy landscape can be explicitly shaped to enforce specific semantic constraints or structural priors without requiring invertible transformations typical of normalizing flows.

Optimization within these latent spaces often employs Langevin dynamics or gradient-based sampling to navigate the energy surface, allowing for the refinement of latent vectors toward high-probability regions corresponding to desired outputs. Unlike standard diffusion processes that rely on fixed noise schedules, latent EBMs dynamically adjust sampling points based on the local energy gradient, enabling more efficient convergence to target configurations. This flexibility supports advanced manipulation tasks, such as attribute editing and conditional generation, where the energy function serves as a differentiable guide to steer the latent trajectory while preserving perceptual fidelity during the subsequent decoding phase.

Despite their advantages, integrating EBMs into latent frameworks introduces challenges related to mode collapse and the stability of gradient estimates in non-convex energy landscapes. Recent advancements address these issues by combining EBMs with flow-matching techniques or utilizing

pre-trained denoising networks to initialize sampling trajectories, thereby enhancing the robustness of the optimization process. Furthermore, the synergy between latent space compression and energy-based regularization allows for scalable modeling of long-range dependencies, making this paradigm particularly effective for high-resolution synthesis and complex video generation tasks where maintaining global coherence is critical.

4.4.3 Real Time Inference and Low Latency Architectures

Achieving real-time inference in generative modeling necessitates a fundamental shift from iterative denoising processes to direct sampling mechanisms. Traditional diffusion models often require hundreds of function evaluations, creating prohibitive latency for interactive applications. Recent advancements in flow matching and rectified flow architectures address this by defining straight probability paths in the latent space, effectively reducing the required inference steps to as few as one or twenty-five evaluations. By modeling the forward process as ordinary differential equations with straight trajectories, these methods minimize the computational burden associated with solving complex stochastic differential equations, thereby enabling high-throughput generation without significant degradation in sample fidelity.

Architectural optimizations further complement algorithmic efficiencies to reduce end-to-end latency. Strategies such as restricting guidance mechanisms to deep network blocks and eliminating per-token multinomial sampling loops have demonstrated superior scalability, particularly when increasing batch sizes beyond thirty-two. Furthermore, the integration of invertible transformations and non-linear independent component estimation allows for efficient density estimation within pre-trained auto-encoding structures. By trading marginal accuracy for substantial speed gains, models like X-MDPT achieve generation times under 1.2 seconds for high-resolution images, while continuous normalizing flows utilizing simplified stacked functions can reduce inference latency to approximately 0.21 seconds through optimized ODE solvers.

Despite these advancements, deploying generative models in strict real-time environments, such as video synthesis or sign language translation, remains challenging due to the inherent computa-

tional cost of high-dimensional data processing. Current video diffusion modules often incur latencies exceeding thirty seconds for short clips, primarily due to the complexity of temporal attention mechanisms and large state spaces. Future low-latency architectures must therefore explore lightweight inductive biases, such as non-local patch aggregation, and joint latent-normalizing flow designs. These approaches aim to decouple quality from computational depth, ensuring that generative systems can meet the stringent timing requirements of practical, interactive deployments while maintaining semantic consistency.

5 Future Directions

Despite the significant theoretical unification of diffusion and flow-based models presented in this survey, several critical limitations persist in current methodologies. The primary gap lies in the computational intractability of exact optimal transport couplings for large-scale datasets, which forces reliance on minibatch approximations that may fail to capture global geometric structures accurately. Furthermore, while high-order solvers and adaptive step-size controllers improve efficiency, they introduce substantial memory overhead and complexity that hinder deployment on resource-constrained edge devices. There is also a notable lack of robustness in manifold extensions, where guidance mechanisms often suffer from numerical instability and Lipschitz explosions near complex data topologies. Finally, the trade-off between strict algebraic reversibility and architectural expressivity remains unresolved, limiting the ability to perform precise latent space inversions without compromising generation quality or training stability.

Future research should prioritize the development of scalable, global optimal transport algorithms that transcend current minibatch limitations, potentially leveraging hierarchical or multi-scale coupling strategies to approximate straight trajectories more faithfully across massive datasets. Concurrently, there is a pressing need to design lightweight, hardware-aware numerical solvers that maintain the accuracy of high-order integration schemes while minimizing memory footprint, perhaps through novel neural architecture searches tailored for specific solver dynamics. Another promising direction involves advancing manifold-aware generative frameworks

that intrinsically embed geometric constraints into the vector field parameterization, thereby preventing off-manifold deviations and ensuring stable guidance without ad-hoc regularization. Additionally, exploring hybrid architectures that dynamically switch between stochastic and deterministic modes based on local trajectory complexity could reconcile the robustness of diffusion training with the speed of flow-based sampling. Investigating self-supervised objectives for learning invertible mappings without explicit Jacobian computation may also unlock new levels of expressivity while preserving efficient inversion capabilities.

The successful realization of these research directions promises to fundamentally transform the landscape of generative modeling by delivering models that are simultaneously fast, stable, and theoretically rigorous. By overcoming the bottlenecks of optimal transport and numerical integration, future systems will enable real-time, high-fidelity image synthesis suitable for interactive applications such as virtual reality, medical imaging diagnostics, and autonomous driving simulation. Moreover, robust manifold extensions will facilitate the reliable generation of data on complex, non-Euclidean domains, opening new frontiers in scientific discovery and 3D content creation. Ultimately, bridging these gaps will establish a new standard for controllable generative AI, where efficiency no longer comes at the expense of quality, and theoretical guarantees translate directly into practical, deployable solutions.

6 Conclusion

This survey has systematically synthesized the converging theoretical and practical landscapes of diffusion and flow-based generative models, establishing that both paradigms share a unified foundation in probability flow ordinary differential equations. We demonstrated that stochastic denoising processes can be rigorously reinterpreted as deterministic velocity regression tasks, where the optimization objectives of score matching and flow matching align under specific conditions. A central finding is the critical role of optimal transport theory in constructing straight generative trajectories; by minimizing kinetic energy and enforcing geodesic paths, models can significantly reduce the integration steps required for high-fidelity sampling without compromising distributional accuracy. Furthermore, our analysis of numerical

solvers highlighted that transitioning from standard first-order Euler methods to high-order adaptive schemes, coupled with strategies to manage algebraic reversibility and manifold constraints, offers a viable pathway to overcome the computational bottlenecks traditionally associated with iterative generative processes. These insights collectively clarify how hybrid architectures can leverage the training stability of diffusion models while inheriting the sampling efficiency inherent to normalizing flows.

The significance of this work lies in its ability to bridge previously disjointed research threads into a coherent framework that guides the development of next-generation controllable image synthesis systems. By explicitly mapping the mathematical correspondences between stochastic and deterministic dynamics, this review dispels the notion that these approaches are mutually exclusive, instead presenting them as points on a continuous spectrum of generative dynamics. This unification provides practitioners with a structured taxonomy for selecting architectural components and optimization strategies tailored to specific efficiency-quality trade-offs. Moreover, the systematic categorization of numerical integration techniques and loss reweighting strategies offers a practical roadmap for mitigating common issues such as numerical stiffness and Lipschitz explosions. Consequently, this survey serves as an essential reference for researchers aiming to design models that are not only theoretically grounded but also scalable for real-time applications in complex domains ranging from medical imaging to 3D content creation.

Looking forward, the field must prioritize the development of robust, geometry-aware solvers that can seamlessly operate on non-Euclidean manifolds while maintaining the computational advantages of straightened flows. Future research should focus on refining minibatch optimal transport approximations to better estimate global couplings without incurring prohibitive costs, thereby further straightening trajectories in high-dimensional latent spaces. Additionally, there is a pressing need to explore adaptive guidance mechanisms that dynamically balance vector field stiffness with generative diversity, ensuring stable convergence even under strict conditional constraints. We call upon the community to move beyond isolated architectural improvements and embrace a holistic design philosophy that in-

tegrates theoretical optimality with practical deployment requirements. By continuing to refine the interplay between transport maps, numerical precision, and geometric consistency, researchers can unlock the full potential of unified generative frameworks, ultimately achieving real-time, high-fidelity synthesis that rivals human creativity.

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